

Models of Curves and Valuations

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Preface

This is a preliminary version of an open book developed as part of the MCLF project.

The text is work in progress. Individual versions should be cited by referring to a specific Git commit or release.

Chapter 1

Valued Fields and Models

1.1 Models of Curves

1.1.1 Basic Definitions

Let K be a field endowed with an additive valuation of rank 1, $v_K : K \rightarrow \hat{\mathbb{R}} := \mathbb{R} \cup \{\infty\}$. Let $\Gamma_K := v_K(K^*) \subset \mathbb{R}$ be the value group of v_K , $\mathcal{O}_K \subset K$ its valuation ring with maximal ideal $\mathfrak{m}_K$ and $k := \mathcal{O}_K/\mathfrak{m}_K$ its residue field. We assume that v_K is discrete, K is complete with respect to v_K and k is perfect. Let X be a smooth, projective and absolutely irreducible curve over K .

Definition 1.1. A *model* $\mathcal{X}$ of X is a proper, flat and normal $\mathcal{O}_K$ -scheme with an identification of its generic fibre $\mathcal{X}_g$ with X .

Example 1.2. Let $K = \mathbb{Q}_p$, $\mathcal{O}_K = \mathbb{Z}_p$ and $X = \mathbb{P}^1(K)$, then $\mathcal{X} = \mathbb{P}^1(\mathcal{O}_K)$ is a model of X with the evident isomorphism from $\mathcal{X}_g = \mathcal{X} \times_{\mathcal{O}_K} K$ to $\mathbb{P}^1(K)$. The special fibre¹ $\mathcal{X}_s$ has one irreducible component which is isomorphic to $\mathbb{P}^1(k)$.

Remark 1.3. The condition of a model being flat is equivalent to it being integral: If $\mathcal{X} \rightarrow \operatorname{Spec} \mathcal{O}_K$ is an integral scheme, then for every $x \in \mathcal{X}$, the stalk $\mathcal{O}_{\mathcal{X},x}$ is an integral ring extension of $\mathcal{O}_K$. Thus $\mathcal{O}_{\mathcal{X},x}$ is a torsion-free module over the discrete valuation ring $\mathcal{O}_K$, hence flat by [Liu02, Corollary 1.2.14]. Conversely, the integrality of a model $\mathcal{X}$ of X follows from its flatness and the integrality of X by [Liu02, Proposition 4.3.8].

Remark 1.4. A point $x \in \mathcal{X}$ of codimension 1 which is regular is normal. This is conversely true by Serre's criterion (see [Liu02, Lemma 8.2.21]).

Example 1.5. Consider the curve X defined by $y^2 = x^3 + px$ over $\mathbb{Q}_p$ with $p \neq 2$. Then

$$\mathcal{X} := \operatorname{Proj} \mathbb{Z}_p[X, Y, Z]/(Y^2Z - X^3 - pXZ^2) \quad (1.1)$$

¹That is, the fibre of the closed point of $\operatorname{Spec} \mathcal{O}_K$.

is a model of X . It is clearly a proper scheme over $\mathcal{O}_K = \mathbb{Z}_p$. For the flatness it suffices to show that $R := \mathbb{Z}_p[X, Y, Z]/(Y^2Z - X^3 - pXZ^2)$ is a torsion-free $\mathbb{Z}_p$ -module by Remark 1.3. But this is true since $Y^2Z - X^3 - pXZ^2$ is irreducible in $\mathbb{Z}_p[X, Y, Z]$.

TBD: Normality

1.1.2 Valuations of Type II

In this section, we assume the valuation v_K to be *normalised*, i.e. $v_K(K^*) = \mathbb{Z}$.

Definition 1.6. Let F_X be a function field of X over K . A valuation $v : F_X \rightarrow \hat{\mathbb{R}}$ is said to be *of type II*, if

1. $v|_K = v_K$,
2. v is discrete, and
3. the extension of residue fields $k(v)|k$ has transcendence degree 1.

A valuation of this type is also known as *geometric valuation*. We write $V_{\text{II}}(F_X)$ for the set of all valuations of type II of F_X . The number

$$m(v) := [\Gamma_v : \Gamma_K] \quad (1.2)$$

is called the *multiplicity of v* , where $\Gamma_v := v(F_X^*)$ is the value group. We say that v is *weakly unramified*, if $m(v) = 1$.

Example 1.7. The *Gauß valuation* on the polynomial ring $K[x]$ is defined as

$$v_g : K[x] \rightarrow \hat{\mathbb{Q}}, \quad v_g \left(\sum_i a_i x^i \right) = \min_i (v_K(a_i)). \quad (1.3)$$

Since $k(v_g) = k(\bar{x})$ (where $\bar{x}$ is the image of x in $k(v_g)$), the Gauß valuation extends naturally and uniquely to $K(x)$, meaning that we have $v_g \in V_{\text{II}}(K(x))$. We will later see that $\{v_g\}$ defines the model $\mathcal{X} = \mathbb{P}^1(\mathbb{Z}_p)$ of $\mathbb{P}^1(\mathbb{Q}_p)$.

Let $\mathcal{X}$ be a model of X . We wish to assign a valuation in $V_{\text{II}}(F_X)$ to each generic point of the special fibre $\mathcal{X}_s$. Suppose that Z is an irreducible component of $\mathcal{X}_s$ with generic point ξ . Since ξ has codimension 1 in $\mathcal{X}$, the stalk $\mathcal{O}_{\mathcal{X}, \xi}$ is a discrete valuation ring. Thus, we can define a valuation on F_X ,

$$v_Z : \text{Frac}(\mathcal{O}_{\mathcal{X}, \xi}) = F_X \rightarrow \hat{\mathbb{Q}}. \quad (1.4)$$

We set

$$V(\mathcal{X}) := \{v_Z : Z \subset \mathcal{X}_s \text{ is an irreducible component}\}. \quad (1.5)$$

Proposition 1.8. *The valuation defined in 1.4 is of type II.*

Proof. Since $\mathcal{X}$ is a flat $\mathcal{O}_K$ -scheme $\mathcal{O}_{\mathcal{X},\xi}$ dominates $\mathcal{O}_K$, and we may scale v_Z such that the restriction of v_Z to K is v_K . Furthermore, we have a closed immersion $Z \hookrightarrow \mathcal{X}$ by definition, inducing a surjection $\mathcal{O}_{\mathcal{X},\xi} \rightarrow \mathcal{O}_{Z,\xi}$. Thus $\text{trdeg}(k(v_Z)) = \text{codim}(\mathcal{O}_{Z,\xi}) = \text{codim}(v_Z) = 1$, showing $v_Z \in V_{\text{II}}(F_X)$. $\square$

Definition 1.9. The *multiplicity* of the irreducible component Z of $\mathcal{X}_s$ containing the generic point ξ is defined as

$$m(Z) := \text{length}(\mathcal{O}_{\mathcal{X},\xi}). \quad (1.6)$$

Proposition 1.10. *It holds $m(v_Z) = m(Z)$.*

Proof. As the model $\mathcal{X}$ is normal, the stalk $\mathcal{O}_{\mathcal{X},\xi}$ is a discrete valuation ring. Let π_ξ denote an uniformizing parameter, then

$$\mathcal{O}_{\mathcal{X}_s,\xi} = \mathcal{O}_{\mathcal{X},\xi}/(\pi_\xi). \quad (1.7)$$

Thus, the length of the stalk at ξ coincides with the length of the chain of ideals generated by the uniformizing parameter of the quotient ring: Since $(\pi_\xi) = (\pi_{Z,\xi}^m)$, where $m = (\Gamma_{v_Z} : \Gamma_K)$, it holds $m(v_Z) = \text{length}(\mathcal{O}_{\mathcal{X},\xi})$ as desired. $\square$

Definition 1.11. A *modification* of a model of X is a morphism $\mathcal{X}' \rightarrow \mathcal{X}$ of $\mathcal{O}_K$ -schemes which induces the identity on the generic fibre.

Remark 1.12. Since a model is proper over $\mathcal{O}_K$, a modification is a proper morphism. Thus since it is dominant, it is also surjective.

The main goal of the first two sections of this chapter is to establish the following theorem.

Theorem 1.13. *The map*

$$\mathcal{X} \mapsto V(\mathcal{X}). \quad (1.8)$$

defines a monotone bijection between

- *the set of isomorphism classes of models of X , and*
- *the set of finite, nonempty subsets of $V_{\text{II}}(F_X)$.*

In the remainder of this section, we prove injectivity of this map.

Lemma 1.14 ([R14, Lemma 3.6]). *Let $\mathcal{X}$ be a model of X and let $\mathcal{X}'$ be a model of another irreducible smooth projective curve X' . Let $\varphi : \mathcal{X}' \rightarrow \mathcal{X}$ be a morphism of $\mathcal{O}_K$ -schemes which induces a cover $X' \rightarrow X$ on the generic fibre. Let η be a generic point of $\mathcal{X}_s$, then the preimage of η under φ is non-empty and consists of generic points of $\mathcal{X}'_s$. If φ is birational, then the preimage consists of precisely one generic point.*

Proof. Let $Z \subset \mathcal{X}_s$ be the irreducible component that contains η . Since φ is surjective (cf. Remark 1.12), there is a point η' on an irreducible component $Z' \subset \mathcal{X}'_s$ which is mapped to η . Since φ is dominant, the induced ring homomorphism $\mathcal{O}_{\mathcal{X},\eta} \rightarrow \mathcal{O}_{\mathcal{X}',\eta'}$ is injective. This is an extension of discrete valuation rings and thus $\eta' \in Z'$ must be generic. In particular, if φ is birational, then both rings are integrally closed in the same field of fractions which makes them equal. Therefore, η' must be unique since $\mathcal{X}'$ is separated. $\square$

Lemma 1.15 ([Ri4, Lemma 3.7]). *Let $\varphi : \mathcal{X}' \rightarrow \mathcal{X}$ be a modification of models of X . Then $V(\mathcal{X}) \subset V(\mathcal{X}')$.*

Proof. This is a consequence of Proposition 1.8 and Lemma 1.14. $\square$

We have a partial order of models of X given by $\mathcal{X} \leq \mathcal{X}'$ if there exists a modification $\mathcal{X}' \rightarrow \mathcal{X}$. By the previous lemma, this gives a partial order on the corresponding finite subsets of $V_{\text{II}}(F_X)$ via inclusion maps.

Lemma 1.16 ([Ri4, Lemma 3.8]). *Let $\mathcal{X}$ and $\mathcal{X}'$ be models of X with $V(\mathcal{X}) \subset V(\mathcal{X}')$. Then there is a modification $\mathcal{X}' \rightarrow \mathcal{X}$.*

Sketch of the proof. TBD. $\square$

Proposition 1.17 ([Ri4, Proposition 3.9], [Liu02, Corollary 8.3.23]). *Two models $\mathcal{X}$ and $\mathcal{X}'$ of X are isomorphic as models of X if and only if $V(\mathcal{X}) = V(\mathcal{X}')$. In particular, the map defined in (1.8) is injective.*

Proof. Suppose that $V(\mathcal{X}) = V(\mathcal{X}')$. Then there exist morphisms $\mathcal{X} \rightarrow \mathcal{X}'$ and $\mathcal{X}' \rightarrow \mathcal{X}$. The composition of these morphisms belongs to the equivalence class of the identity morphism, when considered as a rational map. $\square$

Example 1.18. We continue with Example 1.5 of the model

$$\mathcal{X} = \text{Proj } \mathbb{Z}_p[X, Y, Z]/(Y^2Z - X^3 - pXZ^2)$$

of the elliptic curve $X : y^2 = x^3 + px$ defined over $\mathbb{Q}_p$. We have already checked that it is a model. Let Z be the irreducible component in $\mathcal{X}_s$ containing the generic point ξ and denote the corresponding valuation by v_Z . Z is defined by $\bar{X}_p : y^2 = x^3$, thus $Z = \mathcal{X}_s$ and $V(\mathcal{X}) = \{v_Z\}$.

We want to describe v_Z as

$$v_Z|_{F_X} = v_g \quad \text{Why do we need to restrict?}$$

with respect to the x -coordinate. Using Lemma (In the third talk), we see that $v_Z(p) = 1$ and $v_Z(x) = 0$, since the horizontal divisor on $x = 0$ intersects $\bar{X}_p$ at a unique point

1.2 The Proof of Theorem 1.13

For the proof of Theorem 1.13, it remains to show the surjectivity of the map ((1.8)), i.e., for a given finite, non-empty set of valuations $V \subset V_{\text{II}}(F_X)$, there is a model $\mathcal{X}$ of X such that $V = V(\mathcal{X})$. We will prove this in two steps, which we state in the following proposition.

Proposition 1.19 ([Rĭ4, 3.14, 3.16], [Liu02, 8.3.36]). (i) *Given model $\mathcal{X}$ and a valuation $v \in V_{\text{II}}(F_X)$, there exists a modification $\varphi: \mathcal{X}' \rightarrow \mathcal{X}$ such that $v \in V(\mathcal{X}')$.*

(ii) *Given a model $\mathcal{X}$ and a nonempty subset $V \subset V(\mathcal{X})$, there exists a modification $\varphi: \mathcal{X} \rightarrow \mathcal{X}'$ such that $V = V(\mathcal{X}')$.*

After proving Proposition 1.19, the proof is complete: Starting with a model $\mathcal{X}$ of X , we can apply part (i) for every $v \in V \subset V_{\text{II}}(F_X)$ to obtain a new model that induces all valuations in V , but possibly more. We then apply part (ii) to construct a model that induces exactly the valuations in V .

1.2.1 Proof of the First Part

We fix a model $\mathcal{X}$ of X and a type II valuation $v \in V_{\text{II}}(F_X)$.

Definition 1.20. A point $x \in \mathcal{X}$ is called a *centre* of v if there exists a morphism $f: \text{Spec } \mathcal{O}_v \rightarrow \mathcal{X}$ of $\mathcal{O}_K$ -schemes which maps the closed point to x and makes the following diagram commute:

$$\begin{array}{ccc} \text{Spec } \mathcal{O}_v & \xrightarrow{f} & \mathcal{X} \\ & \nwarrow \quad \nearrow & \\ & \text{Spec } F_X & \end{array}$$

Here, the left map is induced by the inclusion $\mathcal{O}_v \hookrightarrow \text{Frac } \mathcal{O}_v = F_X$, and the right map comes from the identification of X with the special fibre $\mathcal{X}_s$. Specifically, it is the canonical map $\text{Spec } F_X \cong \text{Spec } \mathcal{O}_{\mathcal{X}, \xi} \rightarrow \mathcal{X}$, where ξ is the generic point of $\mathcal{X}$.

Lemma 1.21. *There exists a unique centre $x \in \mathcal{X}$ of v .*

Proof. By the valuative criterion for properness, such a centre exists, and the morphism f is unique (see also [Aut26, Section 01J5, Lemma 01J6]):

$$\begin{array}{ccc} \text{Spec } F_X & \xrightarrow{\quad} & \mathcal{X} \\ \downarrow & \searrow f & \downarrow \\ \text{Spec } \mathcal{O}_v & \xrightarrow{\quad} & \text{Spec } \mathcal{O}_K. \end{array}$$

□

Lemma 1.22 ([Rĭ4, 3.11]). *We have $v \in V(\mathcal{X})$ if and only if the centre x of v is of codimension one. In that case x is the generic point on the special fibre which induces v .*

Proof. Assume $v \in V(\mathcal{X})$. Then v is induced by a generic point $\eta \in \mathcal{X}_s$ on the special fibre. The canonical morphism $\mathrm{Spec} \mathcal{O}_{\mathcal{X},\eta} \rightarrow \mathcal{X}$ shows that v has centre η , which is indeed a point of codimension one. Conversely, suppose $v \in V_{\mathrm{II}}(F_X)$ has center $x \in \mathcal{X}$ of codimension one. The morphism $\mathrm{Spec} \mathcal{O}_v \rightarrow \mathcal{X}$ uniquely factors through $\mathrm{Spec} \mathcal{O}_v \rightarrow \mathrm{Spec} \mathcal{O}_{\mathcal{X},x}$, so $\mathcal{O}_v$ dominates $\mathcal{O}_{\mathcal{X},x}$. Since these are both valuation rings with common fraction field K , it follows that they are in fact equal. Thus $v \in V(\mathcal{X})$. $\square$

Lemma 1.23 ([Rĭ4, 3.12, 3.13]). *Let $x \in \mathcal{X}$ be the centre of v , and let $f \in \mathcal{O}_v^* \subset F_X$ be such that the reduction $\bar{f} \in k(v)$ is transcendental over k . Then:*

- (a) *If $f \in \mathcal{O}_{\mathcal{X},x}$, then $v \in V(\mathcal{X})$*
- (b) *If the rational map $\mathcal{X} \dashrightarrow \mathbb{P}^1(\mathcal{O}_K)$ induced by f is defined at the center x of v , then $v \in V(\mathcal{X})$.*

Proof. (a) By Lemma 1.22, it suffices to show that x is a point of codimension one. Choose an open affine set $\mathrm{Spec} A = U \subset \mathcal{X}$ such that $x \in U$ and $f \in A$. Denote by $\mathfrak{p} \subset A$ the prime ideal corresponding to x . Since v is non-trivial, the center x is not the generic point of $\mathcal{X}$, and so the codimension is not zero. If the dimension of $A/\mathfrak{p}$ is at least one, then the inequality

$$\dim(A_{\mathfrak{p}}) + \dim(A/\mathfrak{p}) \leq \dim(A) = 2$$

implies

$$\mathrm{codim}(\overline{\{x\}}, \mathcal{X}) = \dim(A_{\mathfrak{p}}) \leq 2 - \dim(A/\mathfrak{p}) \leq 2 - 1 = 1,$$

and hence equals one. Thus, it suffices to show that $A/\mathfrak{p}$ is not a field. If $A/\mathfrak{p}$ were a field, it would be a finite extension of k , but this is impossible since $\bar{f} \in A/\mathfrak{p} \subset k(v)$ is transcendental over k .

- (b) It suffices to show, using (a), that $f \in \mathcal{O}_{\mathcal{X},x}$. Let $P = V(f^{-1})$ and $Z = V(f)$ be the sets of poles and zeros of f , respectively. Then the rational map $\mathcal{X} \dashrightarrow \mathbb{P}^1(\mathcal{O}_K)$ is obtained by gluing the morphisms

$$\mathcal{X} \setminus P \rightarrow \mathrm{Spec} \mathcal{O}_K[t] \subset \mathbb{P}^1(\mathcal{O}_K), \quad \text{and} \quad \mathcal{X} \setminus Z \rightarrow \mathrm{Spec} \mathcal{O}_K[t^{-1}] \subset \mathbb{P}^1(\mathcal{O}_K),$$

which are induced by

$$\begin{array}{ccc} \mathcal{O}_K[t] \rightarrow \Gamma(\mathcal{X} \setminus P, \mathcal{O}_{\mathcal{X}}), & \text{and} & \mathcal{O}_K[t^{-1}] \rightarrow \Gamma(\mathcal{X} \setminus Z, \mathcal{O}_{\mathcal{X}}), \\ t \mapsto f & & t^{-1} \mapsto f^{-1} \end{array}$$

and is defined on $U := (\mathcal{X} \setminus P) \cup (\mathcal{X} \setminus Z) = \mathcal{X} \setminus (P \cap Z)$. By assumption, $x \in U$. If $x \in \mathcal{X} \setminus P$, the claim follows immediately, as

$$f \in \Gamma(\mathcal{X} \setminus P, \mathcal{O}_{\mathcal{X}}) \subset \mathcal{O}_{\mathcal{X},x}.$$

If $x \in \mathcal{X} \setminus Z$, then

$$f^{-1} \in \Gamma(\mathcal{X} \setminus Z, \mathcal{O}_{\mathcal{X}}) \subset \mathcal{O}_{\mathcal{X},x}.$$

The image of f^{-1} under the local ring homomorphism $\mathcal{O}_{\mathcal{X},x} \rightarrow \mathcal{O}_v$ is a unit by assumption, hence f^{-1} is also a unit in $\mathcal{O}_{\mathcal{X},x}$. Consequently $f \in \mathcal{O}_{\mathcal{X},x}$. □

We can now finish the proof of Proposition 1.19 i.

Proof of Proposition 1.19 i. Let $f \in \mathcal{O}_v^\times \subset F_X$ be such that the reduction $\bar{f} \in k(v)$ is transcendental over k . We also write $f: \mathcal{X} \dashrightarrow \mathbb{P}^1(\mathcal{O}_K)$ for the rational map induced by f , and let U be its domain of definition. Define $\Gamma_f \subset \mathcal{X} \times_{\mathcal{O}_K} \mathbb{P}^1(\mathcal{O}_K)$ to be the graph of f . Recall that it is defined to be the closure of the graph $f|_U$ in $\mathcal{X} \times_{\mathcal{O}_K} \mathbb{P}^1(\mathcal{O}_K) \supset U \times_{\mathcal{O}_K} \mathbb{P}^1(\mathcal{O}_K)$. This is almost a model of X , as it satisfies all the properties of a model except that it might not be normal.

$$\begin{array}{ccccc} & & \mathcal{X}' & & \\ & & \downarrow & & \\ \mathcal{X} \times_{\mathcal{O}_K} \mathbb{P}^1(\mathcal{O}_K) & \supseteq & \Gamma_f & \longrightarrow & \mathbb{P}^1(\mathcal{O}_K) \\ & & \downarrow & \nearrow f & \\ & & \mathcal{X} & & \end{array}$$

The first projection $\mathcal{X} \times_{\mathcal{O}_K} \mathbb{P}^1(\mathcal{O}_K) \rightarrow \mathcal{X}$ induces a birational morphism $\Gamma_f \rightarrow \mathcal{X}$. The second projection induces a morphism $\Gamma_f \rightarrow \mathbb{P}^1(\mathcal{O}_K)$ which extends $f: \Gamma_f \dashrightarrow \mathbb{P}^1(\mathcal{O}_K)$. Let $\mathcal{X}' \rightarrow \Gamma_f$ denote the normalization of Γ_f . This gives a model of $\mathcal{X}'$ for which $f: \mathcal{X}' \rightarrow \mathbb{P}^1(\mathcal{O}_K)$ is a morphism. By Lemma 1.23 b, this implies that $v \in V(\mathcal{X}')$. □

1.2.2 Proof of the Second Part

We fix a model $\mathcal{X}$ of X and a nonempty subset $V \subset V(\mathcal{X})$. Denote by $\mathcal{E}$ the set of irreducible (integral) components of $\mathcal{X}_s$ that induce the valuations in $V(\mathcal{X}) \setminus V$.

Definition 1.24. A model $\mathcal{X}'$ together with a modification $\varphi: \mathcal{X} \rightarrow \mathcal{X}'$ such that for every integral vertical curve E on $\mathcal{X}$, the set $\varphi(E)$ is a point if and only if $E \in \mathcal{E}$ is called a *contraction* of the $E \in \mathcal{E}$.

It can be shown that if a contraction exists, it is unique up to unique isomorphism. Our goal now is to show that a contraction of $\mathcal{E}$ exists. Once this is done, the proof is complete, since for the contraction $\varphi: \mathcal{X} \rightarrow \mathcal{X}'$, we have by construction $V(\mathcal{X}') = V$.

Lemma 1.25 ([Liu02, Lemma 8.3.29, Lemma 8.3.3]). *Let $\mathcal{L}$ be an invertible sheaf on $\mathcal{X}$ generated by global sections $s_0, \dots, s_n$. Consider the morphism $f: \mathcal{X} \rightarrow \mathbb{P}^n(\mathcal{O}_K)$ associated to these sections. Let E be a vertical curve of $\mathcal{X}_s$ with the reduced induced closed subscheme structure. Then $f(E)$ is reduced to a point if and only if $\mathcal{L}|_E \cong \mathcal{O}_E$.*

Proposition 1.26 ([Liu02, Proposition 8.3.30]). *The following are equivalent.*

- (a) *The contraction $\varphi: \mathcal{X} \rightarrow \mathcal{X}'$ of $\mathcal{E}$ exists.*
- (b) *There exists a Cartier divisor D on $\mathcal{X}$ such that $\deg(D|_{\mathcal{X}_g}) > 0$, that $\mathcal{L} \cong \mathcal{O}_{\mathcal{X}}(D)$ is generated by its global sections, and that for any integral vertical curve E , we have $\mathcal{L}|_E \simeq \mathcal{O}_E$ if and only if $E \in \mathcal{E}$.*

Proof. We only sketch (b) $\implies$ (a). For the other implication and details see [Liu02, Proposition 8.3.30].

As $\mathcal{X}_g$ is an integral projective curve over a field, $\mathcal{L}|_{\mathcal{X}_g}$ is an ample divisor ([Liu02, Proposition 7.5.5]). Replacing D by a positive multiple (which does not change the set $\mathcal{E}$), we can suppose that $\mathcal{L}|_{\mathcal{X}_g}$ is very ample.

Let $f: \mathcal{X} \rightarrow \mathbb{P}^n(\mathcal{O}_K)$ be a morphism associated to sections $s_0, \dots, s_n \in \Gamma(\mathcal{X}, \mathcal{L})$ which generate $\mathcal{L}$. Let Z be the closed subset $f(\mathcal{X}) \subset \mathbb{P}^n(\mathcal{O}_K)$ endowed with the reduced (hence integral) scheme structure. Then f induces a dominant morphism $\mathcal{X} \rightarrow Z$ that we still denote by f . This is an isomorphism on the generic fibre because $\mathcal{L}|_{\mathcal{X}_g}$ is very ample. In particular, $f: \mathcal{X} \rightarrow Z$ is birational. Denote by $\pi: \mathcal{X}' \rightarrow Z$ the normalization morphism. As $\mathcal{X}$ is normal, f uniquely factors into $\varphi: \mathcal{X} \rightarrow \mathcal{X}'$ followed by $\pi: \mathcal{X}' \rightarrow Z$.

$$\begin{array}{ccc} & & \mathcal{X}' \\ & \nearrow \varphi & \downarrow \pi \\ \mathcal{X} & \xrightarrow{f} & Z \end{array}$$

Let E be an integral vertical curve on $\mathcal{X}$. Then we have

$$f(E) \text{ is a point} \iff \mathcal{L}|_E \simeq \mathcal{O}_E \iff E \in \mathcal{E}$$

where the first equivalence is a consequence of Lemma 1.25 and the second one is true by assumption. We claim that $\pi: \mathcal{X}' \rightarrow Z$ is a finite morphism. This then completes the proof because $f(E)$ is reduced to a point if and only if $\varphi(E)$ is a point. The morphism f is projective ([Liu02, Corollary 3.3.32(e)]), so $\mathbb{A} \cong f_*\mathcal{O}_{\mathcal{X}}$ is a coherent sheaf on Z ([Liu02, Corollary 5.3.5]).

Furthermore, $\mathbb{A}$ is a sheaf of integrally closed algebras because $\mathcal{X}$ is normal. As a result, the normalization morphism $\pi: \mathcal{X}' \rightarrow Z$ coincides with the canonical morphism $\underline{\text{Spec}} \mathbb{A} \rightarrow Z$ and is a finite morphism ([Liu02, Exercise 5.1.17]). $\square$

For the construction of a suitable Cartier divisor, we furthermore need the following technical lemma.

Lemma 1.27 ([Liu02, Lemma 8.3.32]). *Let D be an effective horizontal Cartier divisor on $\mathcal{X}$, and $\mathcal{L} = \mathcal{O}_{\mathcal{X}}(D)$. Then there exists a $m_0 \geq 1$ such that $\mathcal{L}^{\otimes m}$ is generated by its global sections for every $m \geq m_0$.*

Proposition 1.26 and Lemma 1.27 show that for a contraction morphism to exist, it suffices that there exists an effective Cartier divisor that meets the closed fibres in a suitable way. To construct such a divisor, it is important that the ring $\mathcal{O}_K$ is complete, which implies it is Henselian.

Definition 1.28. Let A be a Noetherian local ring with maximal ideal $\mathfrak{m}$ and residue field $k := A/\mathfrak{m}$. We say that A is Henselian if one of the following equivalent conditions is fulfilled:

- Hensel's Lemma holds: For every monic $f \in A[T]$, and every simple root $a_0 \in k$ of $\bar{f} \in k[T]$, there exists a lift $a \in A$ (so $\bar{a} = a_0$) such that $f(a) = 0$.
- Every finite A -algebra is a direct sum of local A -algebras.

Lemma 1.29 ([Liu02, 8.3.35]). *Let $x \in \mathcal{X}_s$ be a closed point on the special fiber, then the following assertions hold.*

- (a) *There exists an irreducible horizontal Weil divisor Δ containing x .*
- (b) *There exists an effective horizontal Cartier divisor D such that $\mathcal{X}_s \cap D = \{x\}$.*

Proof. (a) Let $\mathfrak{m}_x$ be the maximal ideal of $\mathcal{O}_{\mathcal{X},x}$ and denote by π an uniformizer of $\mathcal{O}_K$. Let $\mathfrak{p}_1, \dots, \mathfrak{p}_n$ be the prime ideals of $\mathcal{O}_{\mathcal{X},x}$ that are minimal among those containing $\pi\mathcal{O}_{\mathcal{X},x}$. As $\dim \mathcal{O}_{\mathcal{X},x} = 2$ ([Liu02, Proposition 8.3.4]), Krull's principal ideal theorem implies that $\mathfrak{m}_x \neq \mathfrak{p}_i$. There therefore exists an

$$f \in \mathfrak{m}_x \setminus (\cup_i \mathfrak{p}_i).$$

Thus f is a regular element and $V(f)$ does not contain any irreducible component of $\text{Spec } \mathcal{O}_{\mathcal{X},x}$.

There exists an affine open subset $W \ni x$ and a regular element $g \in \mathcal{O}_{\mathcal{X}}(W)$ such that $g_x = f$, and that $V(g) \cap W_s = \{x\}$. Let Δ be the closure in $\mathcal{X}$ of an irreducible component of $V(g)$ passing through x . Then Δ is an irreducible horizontal divisor passing through x .

- (b) Let $E := V(g)$ and let W be as in a. Then E can be seen as an effective Cartier divisor on W . Moreover, $E \rightarrow \operatorname{Spec} \mathcal{O}_K$ is quasi-projective and quasi-finite because $\dim E_s < \dim \mathcal{X}_s = 1$. By a corollary of Zariski's Main theorem ([Liu02, Corollary 4.4.8]), there exists an affine open neighborhood U of x such that $E \cap U$ is an open subscheme of a scheme Z that is finite over $\operatorname{Spec} \mathcal{O}_K$.

$$E \cap U \xrightarrow{\text{open emb.}} Z \xrightarrow{\text{finite}} \operatorname{Spec} \mathcal{O}_K$$

Let D be the connected component of Z containing x . Since $\mathcal{O}_K$ is a Henselian local ring, Z is a disjoint union of local schemes. Hence D is local, and consequently $D \subset E \cap U \subset \mathcal{X}$.

Finally, since D is closed in Z , it is finite over $\operatorname{Spec} \mathcal{O}_K$. A finite morphism $D \rightarrow \operatorname{Spec} \mathcal{O}_K$ between affine schemes is projective ([Liu02, Exercise 3.3.22]), and therefore proper ([Liu02, Theorem 3.3.30]). So the morphisms

$$D \rightarrow \mathcal{X} \rightarrow \operatorname{Spec} \mathcal{O}_K \quad \text{and} \quad \mathcal{X} \rightarrow \operatorname{Spec} \mathcal{O}_K$$

are proper, which implies that $D \rightarrow \mathcal{X}$ is also proper ([?, Proposition 3.3.16(e)]). Consequently, D is closed in $\mathcal{X}$ and therefore a Cartier divisor, as required. $\square$

Now we can finally prove Proposition 1.19 ii.

of Proposition 1.19 ii. Let $Z_1, \dots, Z_n$ be the irreducible components of $\mathcal{X}_s$ that induce the valuations in $V \subset V(\mathcal{X})$. By Lemma 1.29 b, there exists an effective horizontal Cartier divisor D_i such that $\mathcal{X}_s \cap D_i$ is a point of Z_i that belongs to none of the components of $\cdot$. Let $D = \sum_{i=1}^n D_i$. By virtue of Lemma 1.27, if necessary replacing D by a multiple, we can even suppose that $\mathcal{L} := \mathcal{O}_{\mathcal{X}}(D)$ is generated by its global sections.

The theorem then results from Proposition 1.26. Indeed, $\mathcal{L}|_E \simeq \mathcal{O}_E$ if $E \in$ since $E \cap D = \emptyset$; if $E \notin$, then E is some Z_i and $\deg \mathcal{L}|_E \geq \deg \mathcal{O}_{\mathcal{X}}(D_i)|_{Z_i} > 0$, and hence $\mathcal{L}|_E \not\simeq \mathcal{O}_E$. $\square$

Remark 1.30. The proof of Proposition 1.19 ii shows that actually all models $\mathcal{X}$ of X are projective. Indeed one just has to choose $V = V(\mathcal{X})$ in the proof to obtain an effective Cartier divisor D which meets all the irreducible components of the fiber $\mathcal{X}_s$. Since $\mathcal{X}_s$ is a projective curve ([Liu02, Exercise 7.5.4]), $\mathcal{O}_{\mathcal{X}}(D)|_{\mathcal{X}_s}$ is ample ([Liu02, Proposition 7.5.5]). It follows from [Liu02, Corollary 5.3.24] that $\mathcal{O}_{\mathcal{X}}(D)$ is ample. So $\mathcal{X} \rightarrow \operatorname{Spec} \mathcal{O}_K$ is projective ([Liu02, Proposition 5.3.6]).

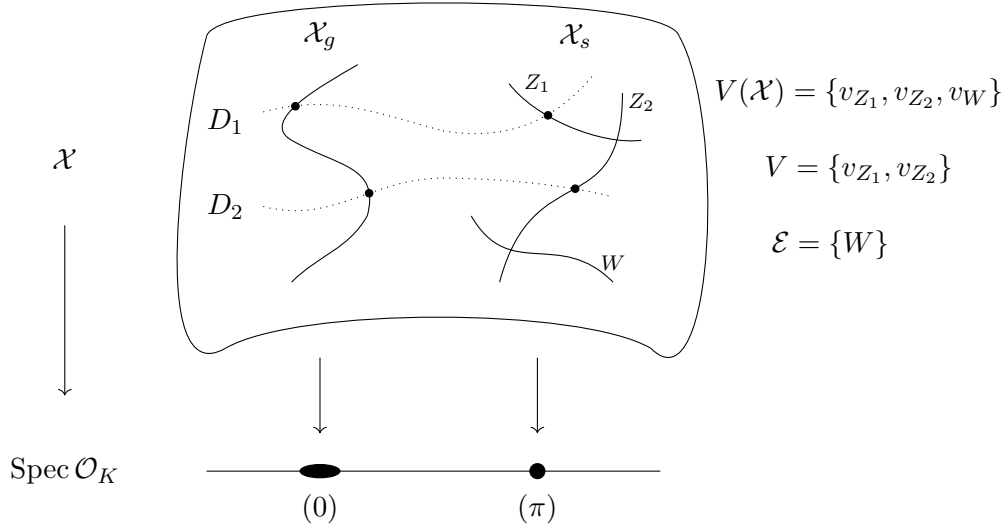


Figure 1.1: Illustration of the proof of Proposition 1.19 ii

1.3 Permanent Models

1.3.1 Models with Reduced Special Fibre

As in the last two sections, we assume X to be a smooth, projective and absolutely irreducible curve over the local field K .

Definition 1.31. A model $\mathcal{X}$ of X is called *permanent*, if $\mathcal{X} \times_{\mathcal{O}_K} \mathcal{O}_L$ is normal for all finite field extensions $L|K$.

Remark 1.32. Since flatness and properness are preserved under base change, a model $\mathcal{X}$ of X is permanent if and only if $\mathcal{X} \times_{\mathcal{O}_K} \mathcal{O}_L$ is a model of $X_L = X \times_K L$.

Definition 1.33. A *pre-model* of X is a flat and proper $\mathcal{O}_K$ -scheme $\mathcal{X}$ with $\mathcal{X} \times_{\mathcal{O}_K} K = X$. That is, $\mathcal{X}$ satisfies all the conditions of a model except of being normal.

Lemma 1.34 ([?, Lemma 1.3]). *Let $\mathcal{X}$ be a pre-model of X . Then $\mathcal{X}$ is normal (i.e. a model) if and only if the following two conditions hold.*

- (i) *The special fibre $\mathcal{X}_s$ has no embedded points, i.e. no point $x \in \mathcal{X}_s$ satisfying $\mathfrak{m}_x = \text{Ann}_R(r)$ for some $r \in R := \mathcal{O}_{\mathcal{X}_s, x}$ (see [Liu02, Definition 7.1.6]).*
- (ii) *The local ring $\mathcal{O}_{\mathcal{X}_\eta}$ is a discrete valuation ring for every generic point $\eta \in \mathcal{X}_s$.*

Proof. According to Serre's criterion for normality [Liu02, Theorem 8.2.23], we need to check the following properties:

(R₁) $\mathcal{X}$ is regular at x when $\text{codim } x \leq 1$, and

(S₂) $\text{depth } \mathcal{O}_{\mathcal{X},x} \geq 2$.

for all $x \in \mathcal{X}$.

The points on $\mathcal{X}$ with codimension 1 are precisely the points on the generic fibre, which is regular by assumption, and the generic point on the special fibre. Since any discrete valuation ring is a local Noetherian domain and vice versa, (R₁) is equivalent to (i). By [Liu02, Proposition 8.2.11], condition (S₂) is equivalent to say that $\text{depth } \mathcal{O}_{\mathcal{X}_s,x} \geq 1$ for all $x \in \mathcal{X}_s$. But this is always true for the generic point, and if x is a different point it is equivalent to say that x is not an embedded point. $\square$

Example 1.35. 1. Consider a pre-model with special fibre $\mathcal{X}_s := \text{Spec } k[x, y]/(x^2, xy)$, which is topologically equivalent to $\mathbb{A}^1(k)$. The point corresponding to the maximal ideal (x, y) is an embedded point.

2. [?, Example 9.8.4].

Proposition 1.36 ([?, Proposition 2.3]). *If $\mathcal{X}_s$ is reduced, then $\mathcal{X}$ is permanent.*

Proof. Let $L|K$ be a finite extension and $l|k$ the extension of the corresponding residue fields. The special fibre of $\mathcal{X}' := \mathcal{X}_{\mathcal{O}_L} = \mathcal{X} \times_{\mathcal{O}_K} \mathcal{O}_L$ is $\mathcal{X}'_s := \mathcal{X}_s \times_k l$, which is reduced since k is perfect by assumption. This means $\mathcal{X}'_s$ satisfies condition (ii) from Lemma 1.34.

For condition (i) let $\eta \in \mathcal{X}'_s$ be any generic point. The stalk $\mathcal{O}_{\mathcal{X}',\eta}$ then is a Noetherian local domain of dimension 1 and it holds

$$\mathcal{O}_{\mathcal{X}'_s,\eta} = \mathcal{O}_{\mathcal{X}',\eta} / \pi_L \mathcal{O}_{\mathcal{X}',\eta}, \quad (1.9)$$

where $\pi_L \in \mathcal{O}_L$ is a uniformising parameter. $\square$

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